

Recall: $\theta_{\mu}^{(\alpha)}(\lambda) := \begin{cases} 0 & \text{if } |\lambda| < |\mu| \\ \binom{|\lambda| - |\mu| + m_1(\mu)}{m_1(\mu)} [p_{\mu-1}^{|\lambda| - |\mu|}] \mathcal{J}_{\lambda}^{(\alpha)} & \text{if } |\lambda| \geq |\mu| \end{cases}$

$\mathcal{S}^{\pi} := (s_1^{\tau_1}, s_2^{\tau_2}, \dots, s_k^{\tau_k})$ for $\mathcal{S} = (s_1 \geq \dots \geq s_k \geq 1)$
 $\pi = (\tau_1, \tau_2, \dots, \tau_k) \in \mathbb{Z}_{\geq 0}^k$

Conjecture: (Lesselle)

$$(-1)^{|\mu|} z_{\mu} \theta_{\mu}^{(\alpha)}(\mathcal{S}^{\pi}) \in \mathbb{Z}_{\geq 0}[b, -s_1, -s_2, \dots, \tau_1, \tau_2, \dots]$$

Let $\gamma = \sqrt{\alpha}^{-1} - \sqrt{\alpha}$,

$$z - (z + \sqrt{\alpha}^{-1} e(\lambda)) \prod_{i=1}^{e(\lambda)} \frac{z + \sqrt{\alpha}^{-1}(i-1) - \sqrt{\alpha} \lambda_i}{z + \sqrt{\alpha}^{-1} i - \sqrt{\alpha} \lambda_i} = \sum_{\ell \geq 0} B_{e,2}^{(\alpha)}(\lambda) z^{-\ell}$$

$$Ch_{\mu}(\lambda) := \alpha^{\frac{e(\mu) - |\mu|}{2}} z_{\mu} \cdot \theta_{\mu}^{(\alpha)}(\lambda)$$

Let $\mathcal{J}_{\mathcal{S}^{\pi}} := \{ \sqrt{\alpha} s_i - \sum_{j=1}^{i-1} \tau_j / \sqrt{\alpha}, i = 1, 2, \dots, k \}$

$$\mathcal{O}_{\mathcal{S}^{\pi}} := \{ \sqrt{\alpha} s_i - \sum_{j=1}^{i-1} \tau_j / \sqrt{\alpha}, i = 1, 2, \dots, k \}$$

Fact: $B_e^{(\alpha)}(\mathcal{S}^{\pi}) = (-1)^{\sum_{i=1}^e \tau_i} e_e[\mathcal{J}_{\mathcal{S}^{\pi}} - \mathcal{O}_{\mathcal{S}^{\pi}}], e \geq 2$

where $p_k[\mathcal{J}_{\mathcal{S}^{\pi}} - \mathcal{O}_{\mathcal{S}^{\pi}}] := \sum_{i \in \mathcal{J}_{\mathcal{S}^{\pi}}} i^k - \sum_{o \in \mathcal{O}_{\mathcal{S}^{\pi}}} o^k$

$p[\mathcal{J}_{\mathcal{S}^{\pi}} - \mathcal{O}_{\mathcal{S}^{\pi}}] := \sum e_p p_p[\mathcal{J}_{\mathcal{S}^{\pi}} - \mathcal{O}_{\mathcal{S}^{\pi}}]$ if $p = \sum e_p \cdot p_p \leftarrow \text{power-sum expansion}$

Let $\ell_1, \ell_2, \dots, \ell_k \in \mathbb{Z}_{\geq 2}$. We have found an explicit combinatorial expansion of

$B_{\ell_1} \cdot B_{\ell_2} \cdots B_{\ell_k}$ as a linear combination of Jack characters Ch_{μ} in terms of **Lukasiewicz ribbon paths**.

Def: **Excursion** Γ of **length** ℓ :

- a lattice path with steps of the form $(1, k)$ $k \in \mathbb{Z}$ that starts at $(0, 0)$, finishes at $(\ell, 0)$ and does not touch the Ox axis

Excursion Γ on k sites of lengths $\ell_1, \ell_2, \dots, \ell_k$:

- an ordered tuple of excursions $\vec{\Gamma} = (\Gamma_1, \dots, \Gamma_k)$ of lengths $\ell_1, \ell_2, \dots, \ell_k$ + possible pairings of a down step of increment s with an up step of the same increment

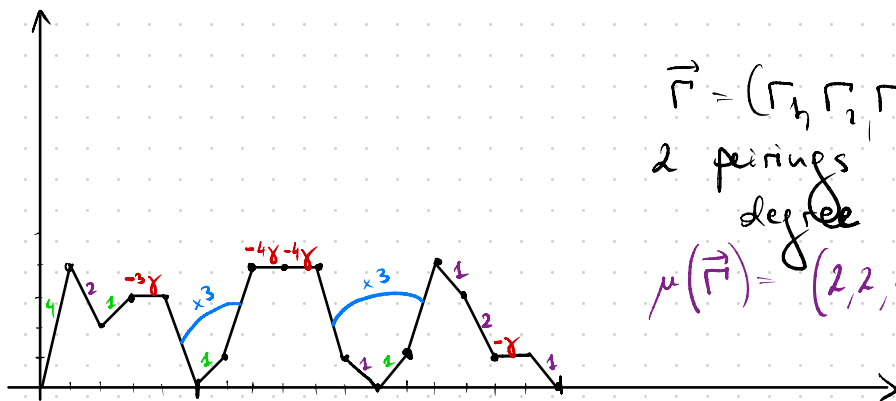
Denote their set by $R(\ell_1, \ell_2, \dots, \ell_k)$

Lukasiewicz ribbon path is a ribbon path such that all the non-paired up steps have increment 1

Denote their set by $L(\ell_1, \ell_2, \dots, \ell_k)$

For $\vec{\Gamma} \in R(\ell_1, \dots, \ell_k)$ let $\mu(\vec{\Gamma})$ be the partition given by the increments of unpaired down-steps in $\vec{\Gamma}$.

Example:



$$\vec{\Gamma} = (\Gamma_1, \Gamma_2, \Gamma_3) \in \mathbb{R}(5, 6, 6)$$

2 pairings of degree 3

$$\mu(\vec{\Gamma}) = (2, 2, 1, 1, 1)$$

$$\text{wt}^{\text{op}}(\vec{\Gamma}) = \prod_{i \geq 1} (-i\gamma)^{\# \rightarrow \text{at height } i} \times \prod_{i \geq 1} p_i^{\# \uparrow \text{ of increment } i} \times \prod_{i \geq 1} (\alpha_i)^{\# (\downarrow \uparrow) \text{ of degree } i} \times \prod_{i \geq 1} \frac{\alpha_i^2}{7p_i}^{\# \downarrow \text{ of increment } i}$$

$$\text{wt}(\vec{\Gamma}) = \prod_{i \geq 1} (-i\gamma)^{\# \rightarrow \text{at height } i} \times \prod_{i \geq 1} (\alpha_i)^{\# (\downarrow \uparrow) \text{ of degree } i}$$

Ex:

$$\frac{\text{wt}^{\text{op}}(\vec{\Gamma})}{\text{wt}(\vec{\Gamma})} = \frac{(-\gamma)(3\gamma)(-4\gamma)^2 (3\alpha)^2 \cdot p_4 \cdot p_1^3 \cdot \alpha^5 \cdot \left(\frac{\alpha^2}{\partial p_2}\right)^2 \cdot \left(\frac{\alpha}{\partial p_1}\right)^3}{(-\gamma)(-3\gamma)(-4\gamma)^2 3^2}$$

Theorem (B.D.D.) Let $e_1, e_2, \dots, e_k \in \mathbb{Z}_{\geq 2}$

Then $B_{e_1} \cdots B_{e_k} = \sum_{\vec{\Gamma} \in \text{ch}(e_1, \dots, e_k)} \text{wt}(\vec{\Gamma}) \cdot \text{Ch}_{\mu(\vec{\Gamma})}$

Corollary 1 Let $\deg(B_e) := e-2, \deg(\gamma) = 1$.

(A) $\text{Ch}_{\mu} \in \mathbb{Z}[\gamma, B_2, B_3, \dots]$ and each monomial has degree $= |\mu| - e(\mu) - 2k \quad k \geq 0$

(B) $\text{Ch}_{\mu} \in \mathbb{Z}[\gamma, -\frac{1}{\alpha} s_1, -\frac{1}{\alpha} s_2, \dots, \frac{1}{\alpha}, \frac{1}{\alpha}, \dots]$

①+③ = integrality in Lisselle's conjecture

Main tool: Nazarov-Sklyanin operators

$$L = \begin{bmatrix} -\gamma & p_{1,1} & p_{1,2} & \dots \\ p_{1,1}^+ & -2\gamma & p_{1,1} & \dots \\ p_{2,1}^+ & p_{1,1}^+ & -3\gamma & \dots \\ \vdots & \vdots & \vdots & \ddots \end{bmatrix}$$

$$P = (p_{i,k})_{k \geq 1}$$

$$p_{i,k} = \sqrt{\alpha}^{-1} \cdot p_k$$

$$P^+ = (p_{k,1}^+)_{k \geq 1}$$

$$p_{k,1}^+ = \sqrt{\alpha} \cdot k \frac{\partial}{\partial p_k}$$

Theorem: (Nazarov-Sklyanin '13)

$$\forall \ell \geq 0 \quad PL^\ell P^+ j_\lambda^{(\alpha)} = B_{\ell+2}^{(\alpha)}(\lambda) \cdot j_\lambda^{(\alpha)}$$

P-f of the combinatorial formula:

$$\textcircled{1} \quad B_{e_1} \dots B_{e_k} = (B_{e_1} \dots B_{e_k} \cdot j_\lambda) \Big|_{\substack{\text{Nazarov-Sklyanin} \\ p_i = \delta_{i,1}}} \left[(PL^{e_1-2} P^+) \dots (PL^{e_k-2} P^+) j_\lambda \right]_{p_i = \delta_{i,1}}$$

② Interpret $(PL^{e_1-2} P^+) \dots (PL^{e_k-2} P^+)$ in terms of paths:

$$(PL^{e_1-2} P^+) \dots (PL^{e_k-2} P^+) = \sqrt{\alpha}^{-e_1 - \dots - e_k} \sum_{\vec{\Gamma} \in R(e_1, \dots, e_k)} \sqrt{\alpha}^{\#\vec{\Gamma}} \text{wt}^\alpha(\vec{\Gamma})$$

The appearances of pairings from the commutation relation $\downarrow \overset{k}{\curvearrowright} \uparrow$ comes from the commutation relation $[\frac{k\partial}{\partial p_k}, p_k] = k$

$$(3) \quad \vec{\Gamma} \in \mathcal{R}(e_1, \dots, e_k) \setminus \mathcal{L}(e_1, \dots, e_k)$$

$$\text{Then } (\text{wt}^{\text{op}}(\vec{\Gamma}) J_{\lambda})|_{p_i = \delta_{i,1}} = 0$$

(4) When $\vec{\Gamma} \in \mathcal{L}(e_1, \dots, e_k)$ and $\mu = \mu(\vec{\Gamma})$ then

$$\left[\prod_{i \geq 1} p_i^{\# \uparrow} \cdot \prod_{i \geq 1} \left(\alpha \frac{\partial}{\partial p_i} \right)^{\# \downarrow} J_{\lambda} \right]_{p_i = \delta_{i,1}} =$$

$$\left[\alpha^{\ell(\mu)} \cdot \prod_{i \geq 1} p_i^{\# \uparrow} \cdot \prod_{i \geq 1} \left(\frac{\partial}{\partial p_i} \right)^{m_i(\mu)} J_{\lambda} \right]_{p_i = \delta_{i,1}} =$$

$$\alpha^{\ell(\mu)} \cdot \binom{|\lambda| - |\mu| + m_1(\mu)}{m_1(\mu)} \cdot z_{\mu} \cdot [p_{\mu}^{\lambda - \mu}] J_{\lambda}$$

$$= \alpha^{\ell(\mu) + |\mu|} \text{Ch}_{\mu}$$

(5) For any $\vec{\Gamma} \in \mathcal{L}(e_1, \dots, e_k)$

$$\ell(\mu(\vec{\Gamma})) + |\mu(\vec{\Gamma})| = e_1 + \dots + e_k - \# \rightarrow - \sum_{k=1}^{\infty} \downarrow \uparrow \quad \square$$

P-f of Corollary 1:

Analysis of the structure of paths that appear in Th 1.

(A) Let μ -partition. Then

$$B_{\mu_1+1} \dots B_{\mu_{\ell}+1} = \text{Ch}_{\mu} + \sum_{\rho \prec_{\text{LEX}} \mu}^{\mathbb{Z}_{\geq 0}[-\infty]} \alpha_{\rho}^{\mu} \cdot \text{Ch}_{\rho}$$

$$\Rightarrow \text{Ch}_{\mu} \in \mathbb{Z}[\gamma, \beta_2, \beta_3, \dots].$$

$$\mathbb{Z}[\gamma][\text{Ch}_\mu : \mu \in Y] = \mathbb{Z}[\gamma, B_2, B_3, \dots] \subseteq S_\alpha^*$$

↑
Shifted symmetric functions

Def: $f \in \mathbb{Q}(\sqrt{\alpha})[s_1, s_2, \dots, s_k]$ is a shifted symmetric polynomial if f is symmetric in $s_i - i/\alpha$. A shifted symmetric function of

degree $n \equiv (f_k)_{k \geq 1}$ s.t.

- $\deg(f_k) = n, k \gg 0$, f_k - shifted sym. pol. in k variables
- $f_{k+1}(s_1, \dots, s_k, 0) = f_k(s_1, \dots, s_k)$

Theorem: (Féray '15) Characterization of Jack characters

Θ_μ is the unique element of S_α^* s.t.

- (1) $\deg(\Theta_\mu) = |\mu|$, $\text{top}(\Theta_\mu) = \alpha^{|\mu| - l(\mu)} \cdot \frac{p_\mu}{z_\mu}$
- (2) $\Theta_\mu(\lambda) = 0$ if $|\lambda| < |\mu|$

Idea: Use it to prove the main combinatorial formula for Θ_μ .